

Understanding Warmup-Stable-Decay Learning Rates: A River Valley Loss Landscape Perspective

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The Challenge: Fixed Compute Budget in LLM Training

Pre-determined Budget



Traditional schedules require a *fixed compute budget*.

Difficult to adapt to new data or compute.

Scaling & Retraining Costs



Schedule isn't a *natural continuation* for all data.

Scaling laws are costly: each budget requires *retraining* to adjust the schedule.



Optimal Model

Solution: Warmup-Stable-Decay (WSD) Schedule

Constant LR Main Branch

Indefinite Continuation

Decaying LR Branches

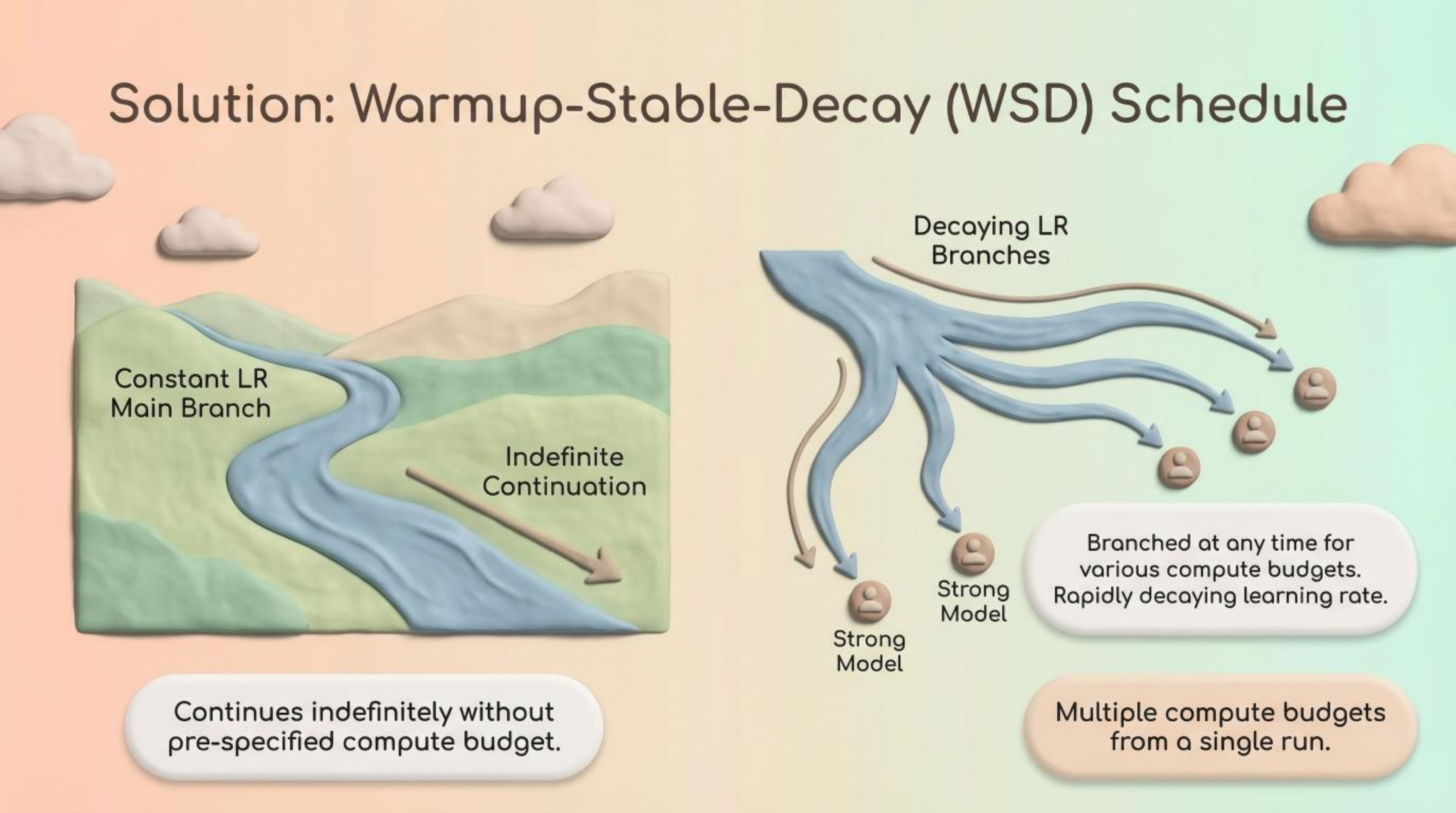
Strong Model

Strong Model

Branched at any time for various compute budgets. Rapidly decaying learning rate.

Multiple compute budgets from a single run.

Continues indefinitely without pre-specified compute budget.



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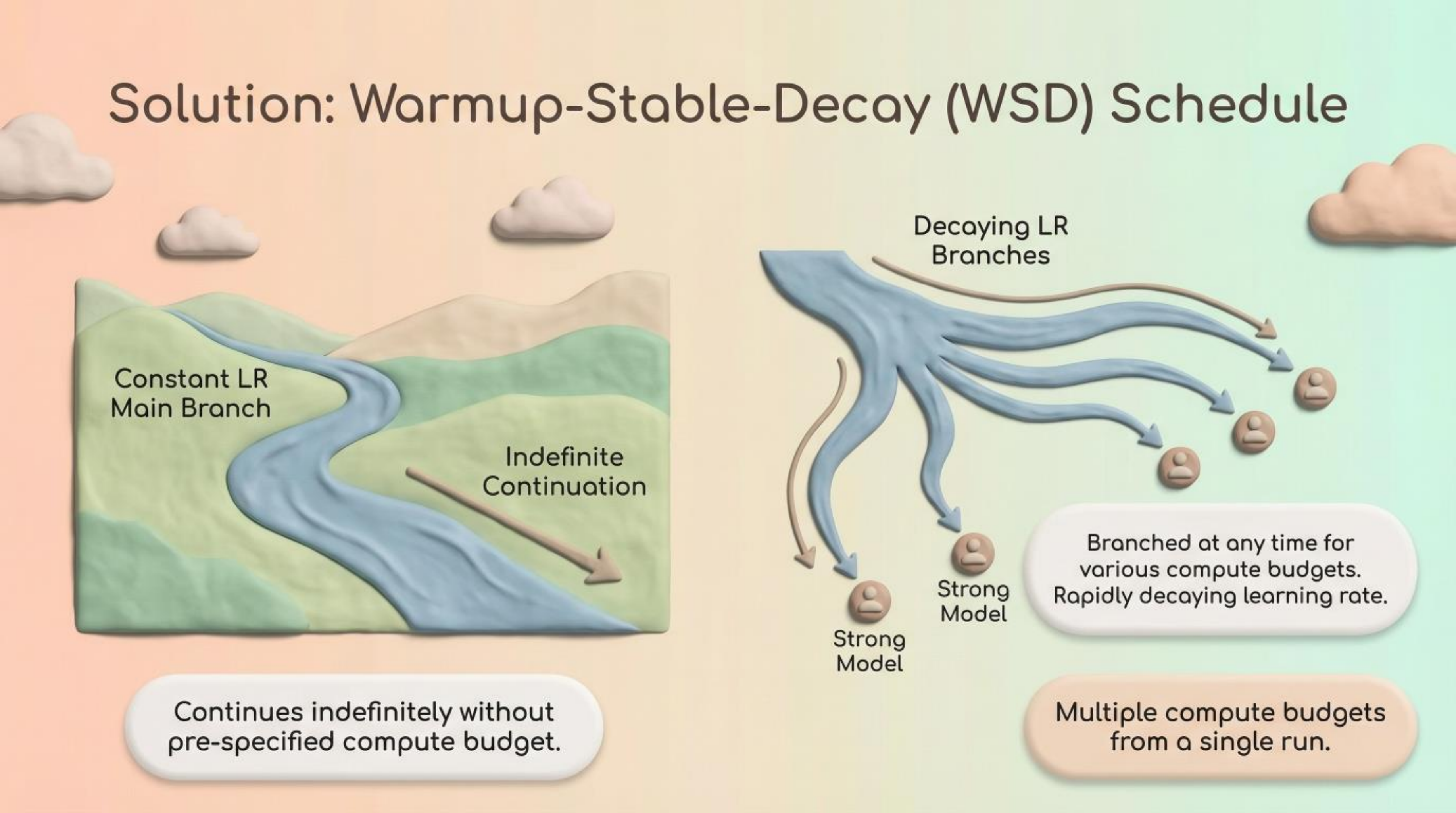
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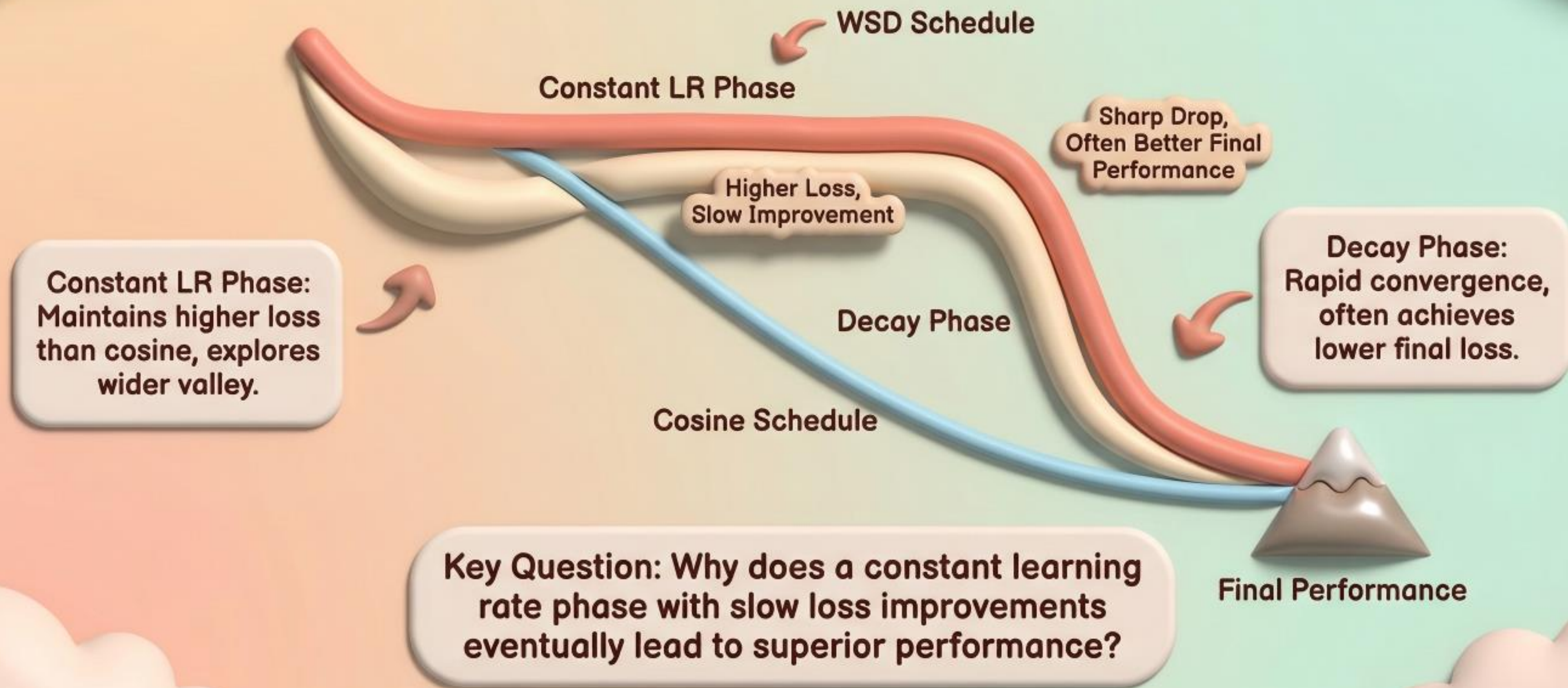
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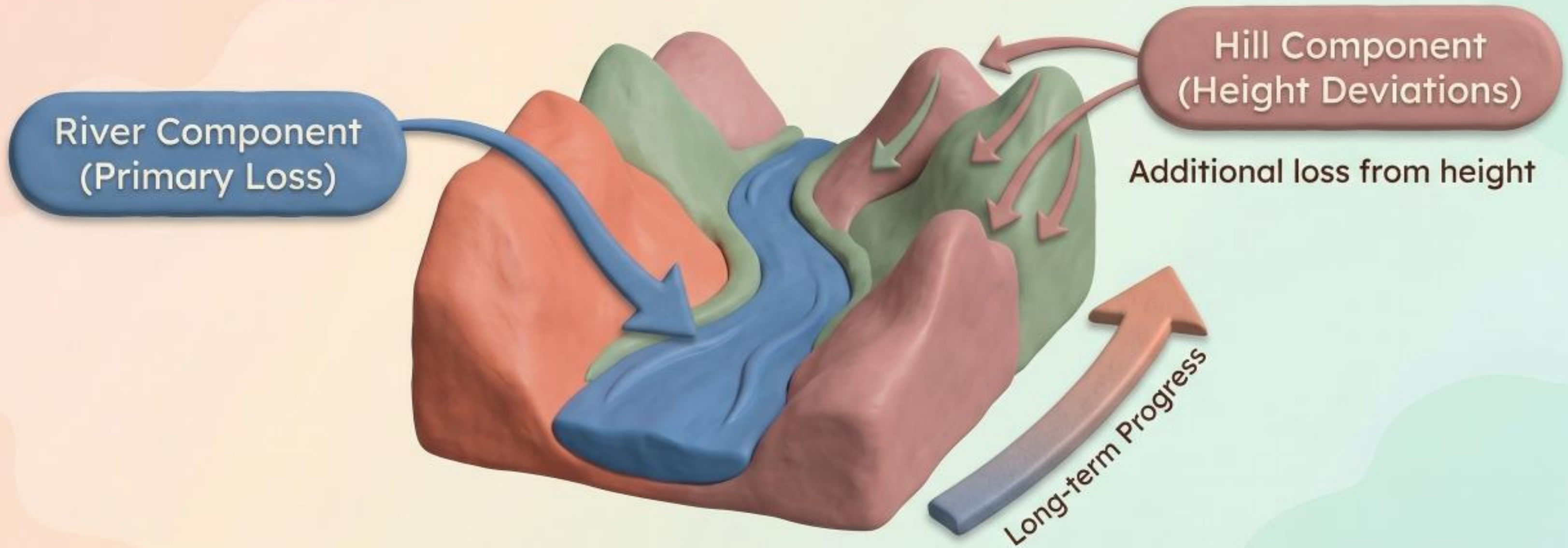
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The Intriguing WSD Loss Curve



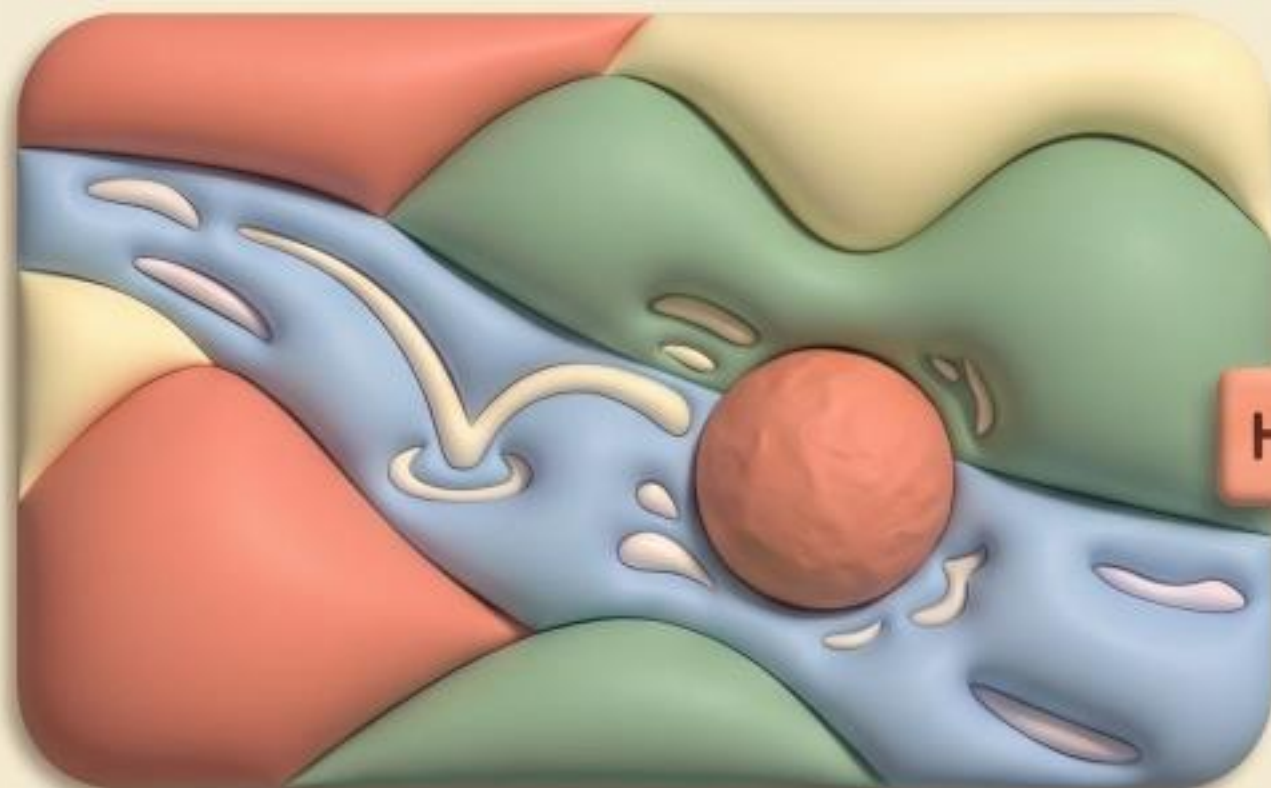
River Valley Landscape: Core Concept



- Loss decomposes into river component (primary) and hill component (additional).
- Progress is primarily determined by the river component in the long run.

Two Key Properties for Effective Optimization

Sustained High Learning Rate



Maintains larger learning rate for longer. Faster progress in river direction, but increased "hill component" loss due to stochastic bouncing.

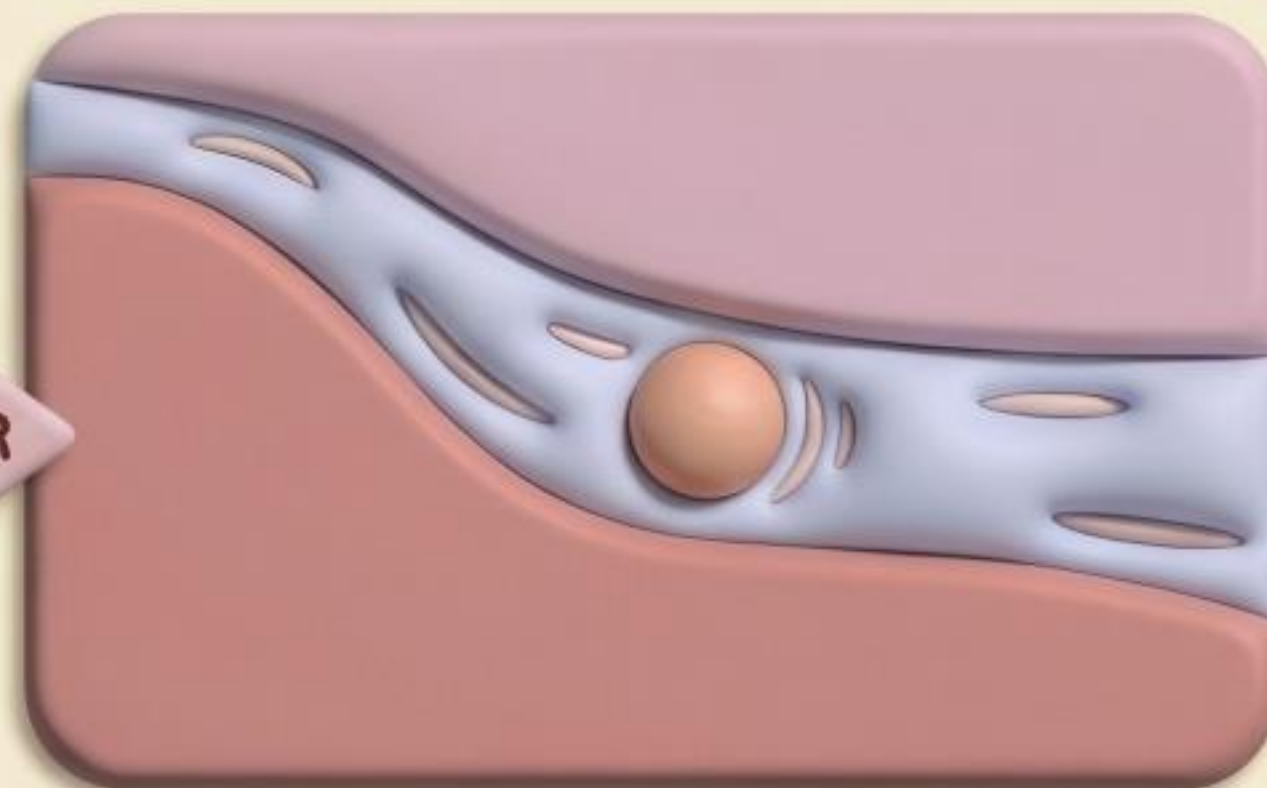
Optimization Flow

High LR

Low LR

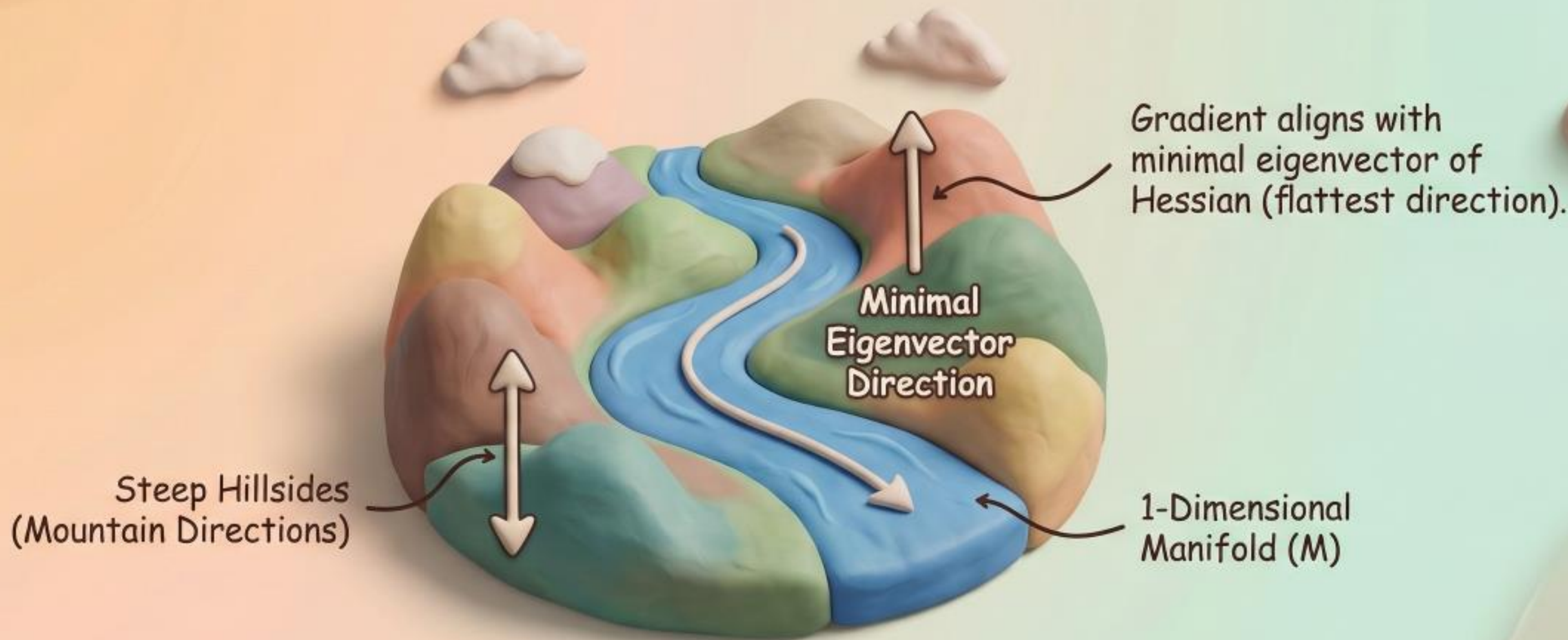
Convergence

Final Low Learning Rate



Essential near completion. Reduces oscillations in mountain direction, ensuring convergence close to the river.

Theoretical Framework: Formal Definition



1. Analyticity:
Smoothness of
loss landscape.



2. Bounded
Hessian: Limits
on curvature.



3. Eigengap
Existence:
Clear separation
of directions.



4. Slow Spinning:
Gradual change in
river direction.

Gradient Flow Dynamics on River Valley

Quick Convergence



Iterates quickly converge to a neighborhood of the river.

Close Tracking

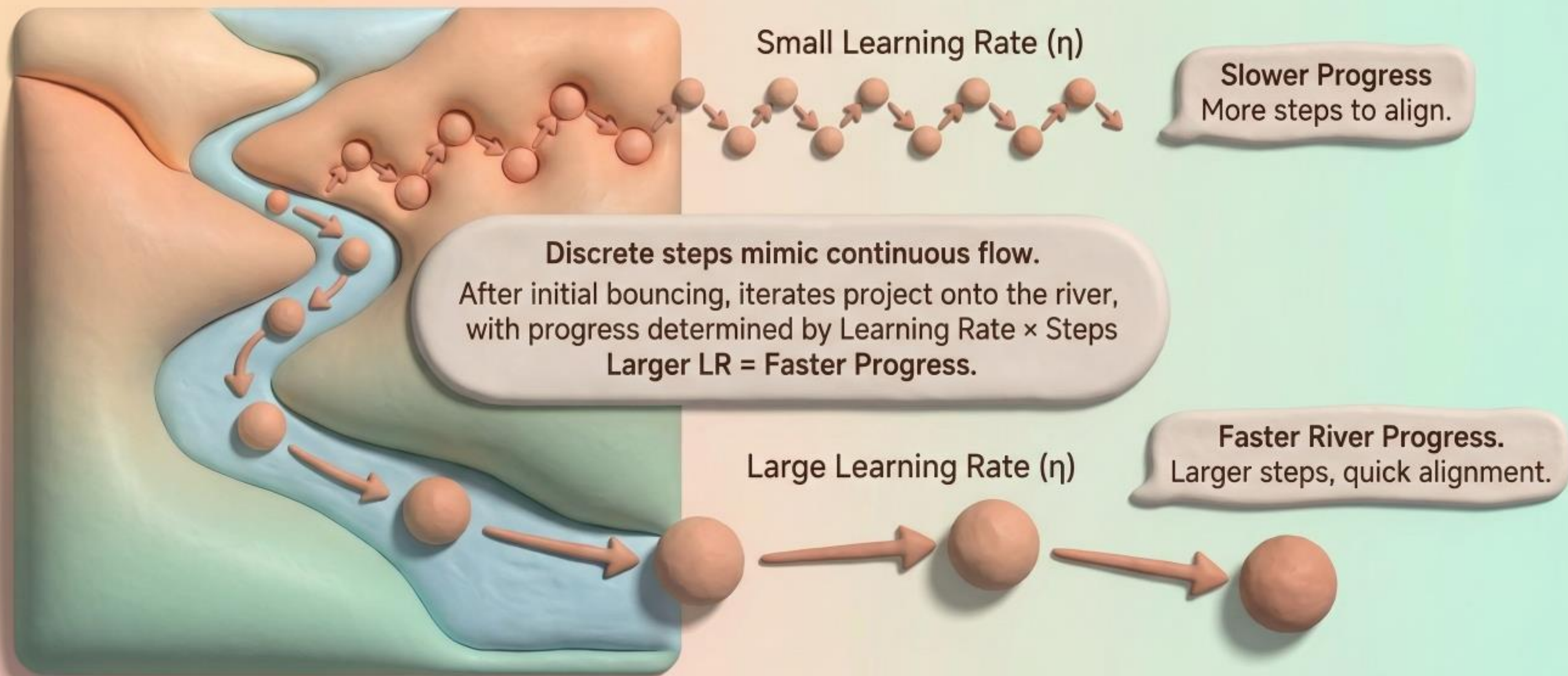


Then tracks the river closely with same pace.



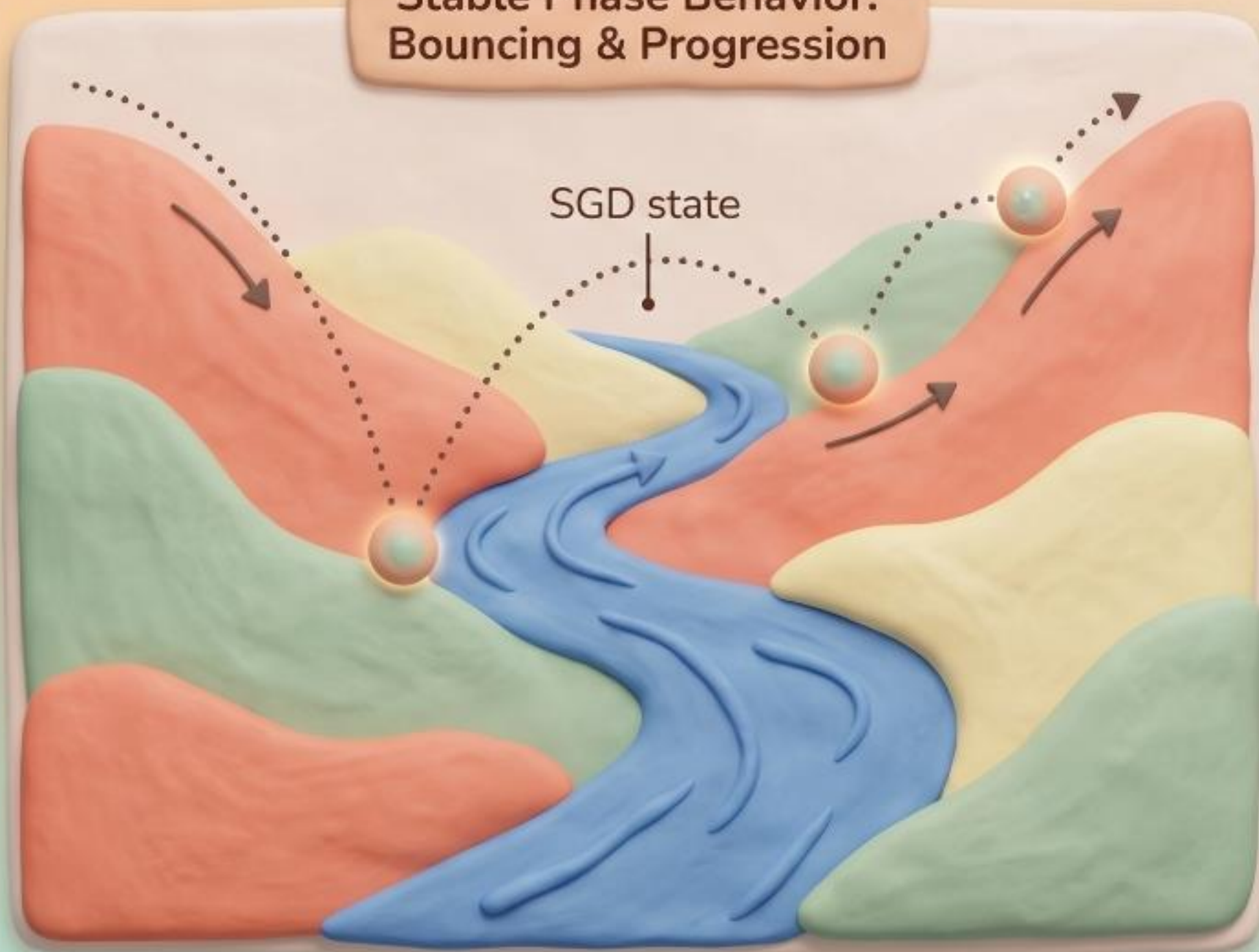
Geometric and time-alignment error vanish when the river spins slowly.

Gradient Descent Dynamics



Stochastic Gradient Descent: Stable Phase

Stable Phase Behavior:
Bouncing & Progression



Expected Loss Decomposition

$$\text{Expected Loss} \approx \text{River Component } (L(x^{(T)})) + \text{Hill Component } \left(\frac{(d-1)\eta\sigma^2}{2} \right)$$



River Component:
Follows Reference
Flow

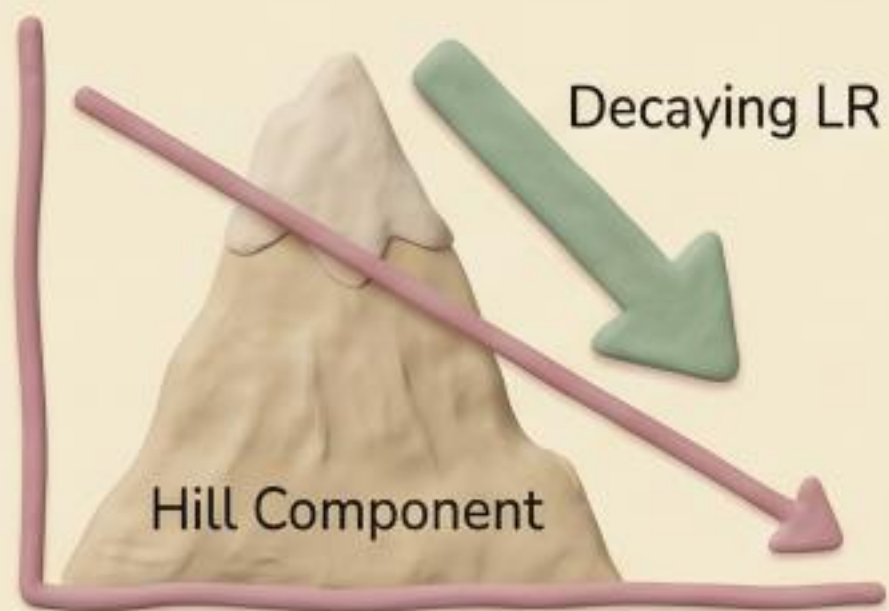


Hill Component:
Increases linearly
with LR (η)

Constant LR η causes bouncing due to noise, preventing convergence to the river while progressing along it.

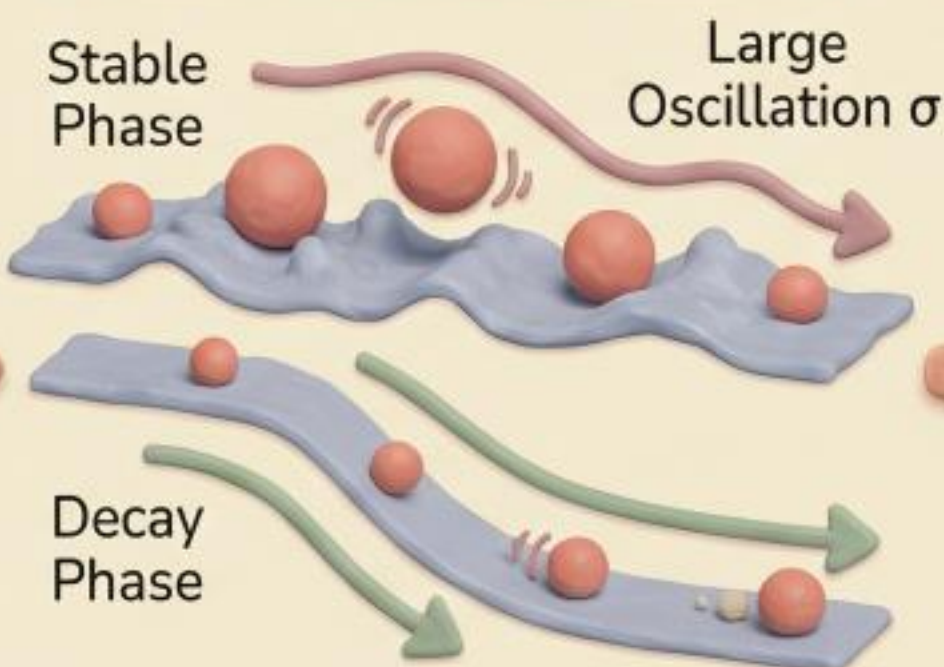
Stochastic Gradient Descent: Decay Phase

Rapid Hill Reduction



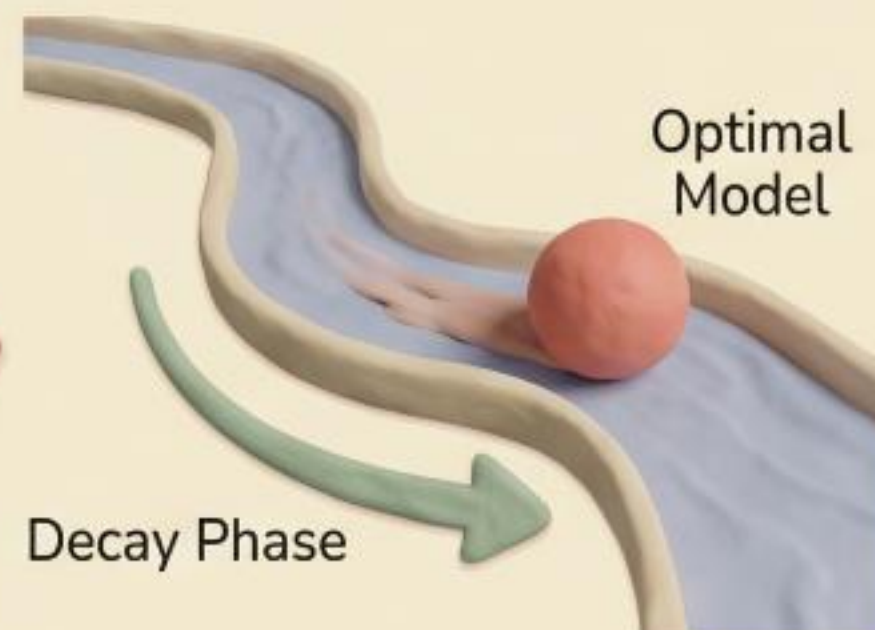
Proper decaying schedule rapidly reduces hill component of loss, decreasing linearly with decaying learning rate.

Faster Loss Decrease (High Oscillation)



When oscillation σ is large, loss decrease appears faster in decay phase than stable phase.

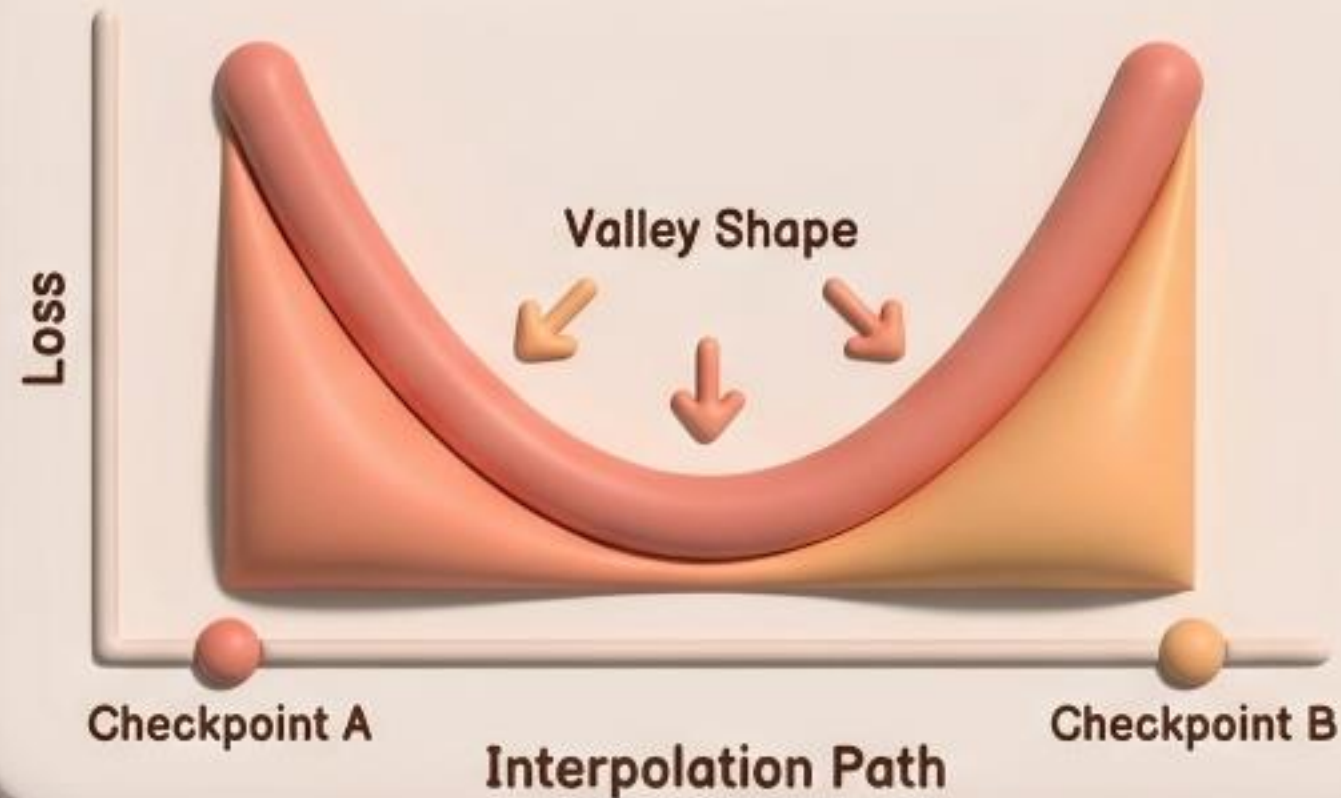
Progress Along River



Decaying phase also makes significant progress along the river component of loss.

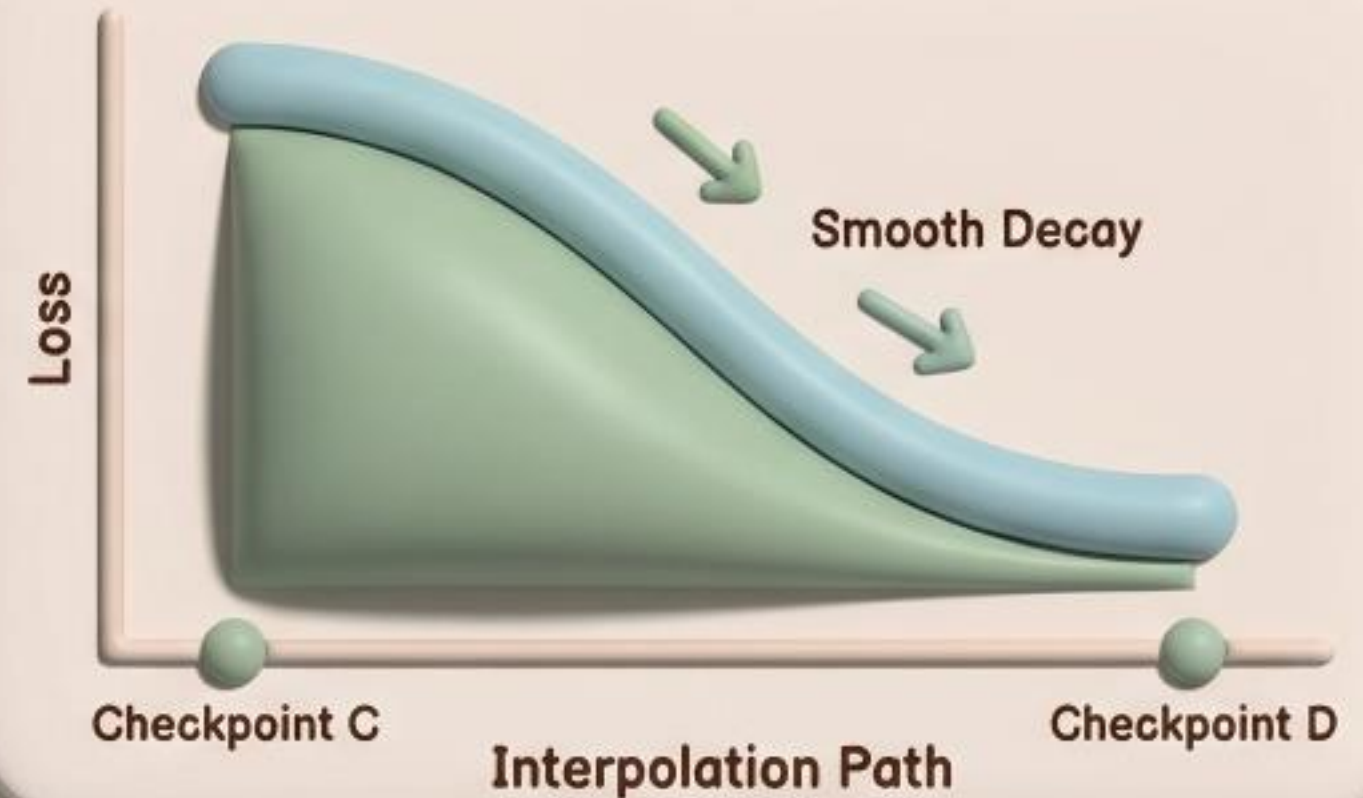
Visualizing the River Valley: Empirical Evidence

Stable Phase: Convex "Valley"



Exhibits convex unimodal shape resembling a valley.

Decay Phase: Smooth Monotone Decay



Shows smooth monotone decay.

Conclusion: These observations validate the river valley analogy in GPT-2 pretraining experiments on OpenWebText.

What Shapes the River Valley: Token Uncertainty

Deterministic Tokens

- Contribute to "river" direction
- Create flatter regions

Highly Deterministic Tokens
(Facts, Knowledge)
→ Flatter Regions

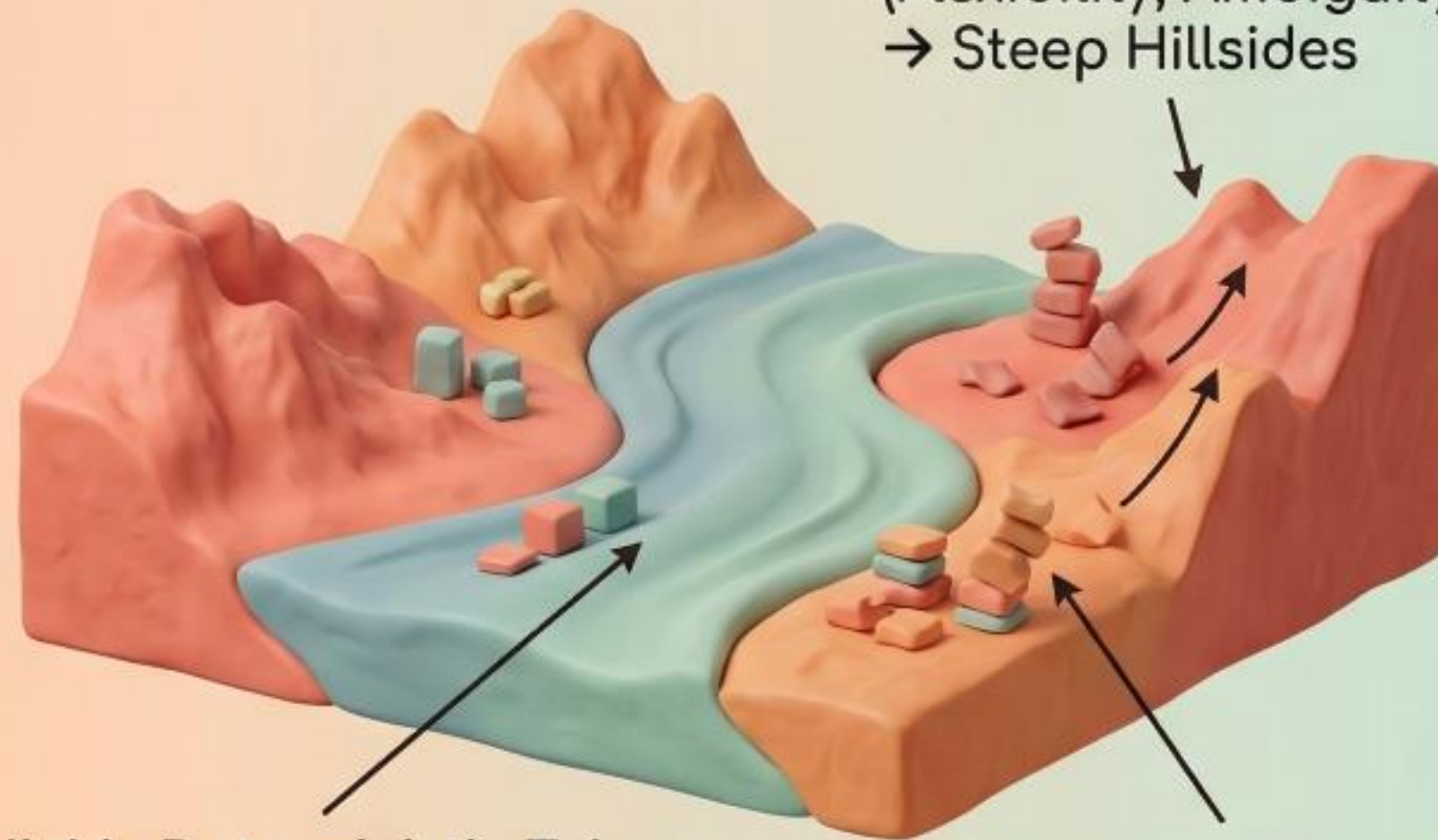
Uncertain Tokens
(Flexibility, Ambiguity)
→ Steep Hillsides

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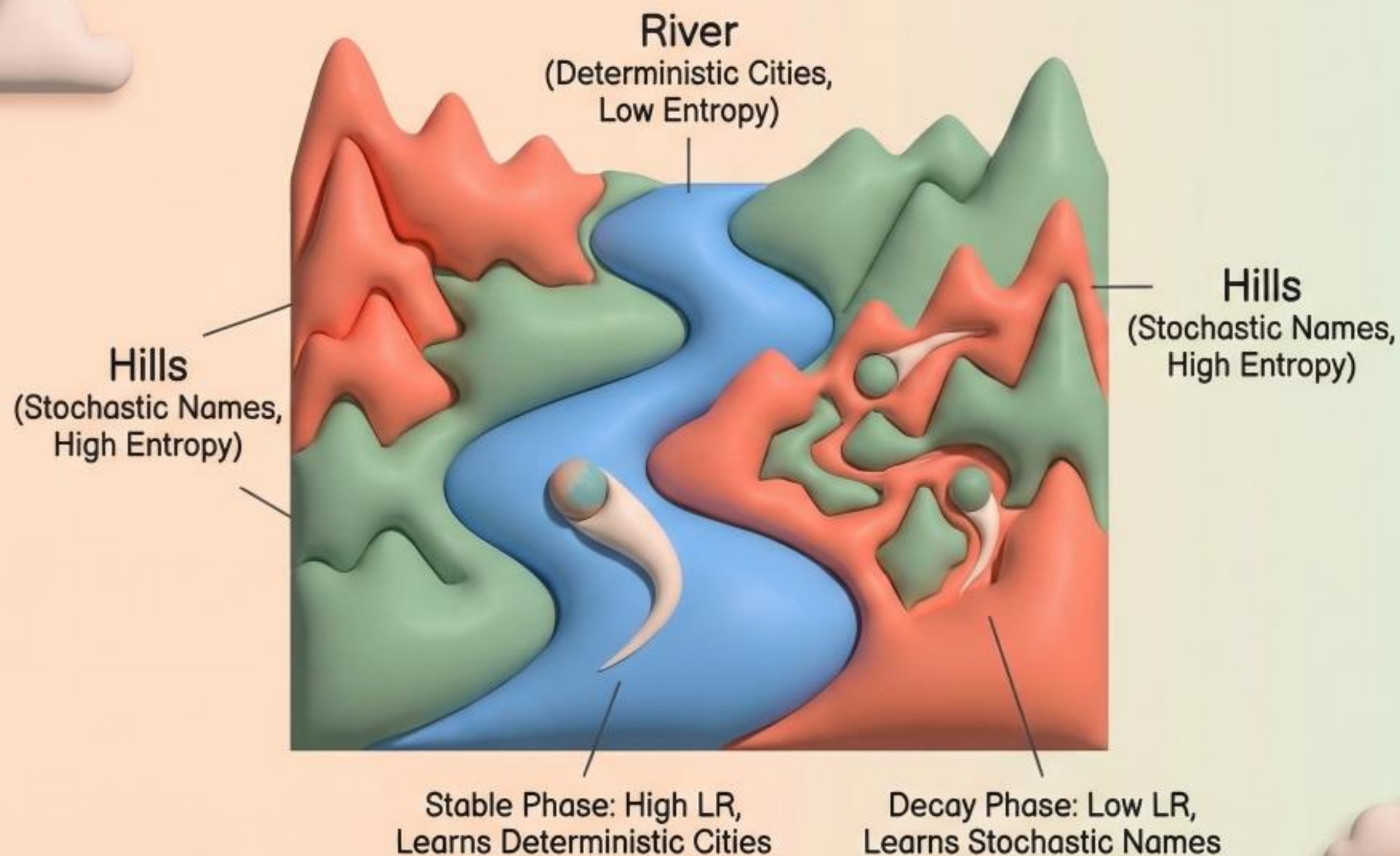
Uncertain Tokens

- Form steep hillsides
- Increase loss landscape complexity

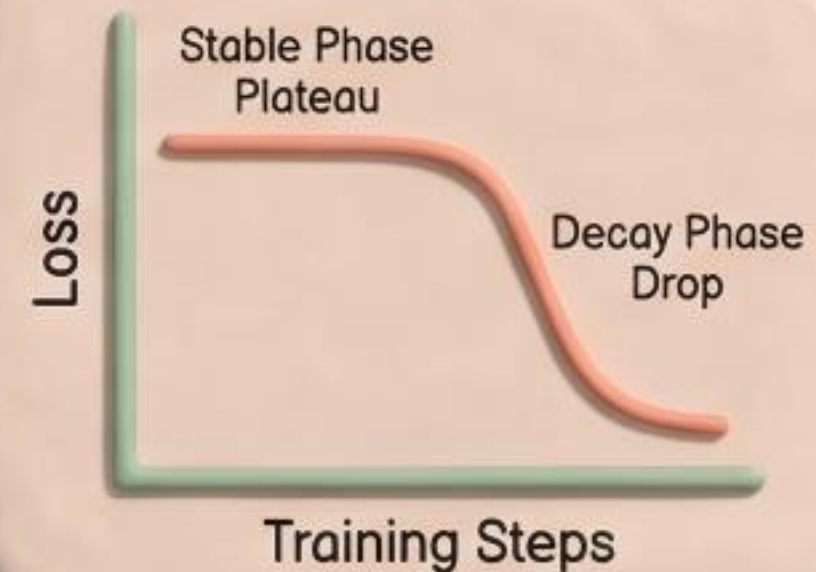
This heterogeneity in stochasticity naturally leads to river valley structure.



Toy Bigram Language Model



WSD Loss Curve



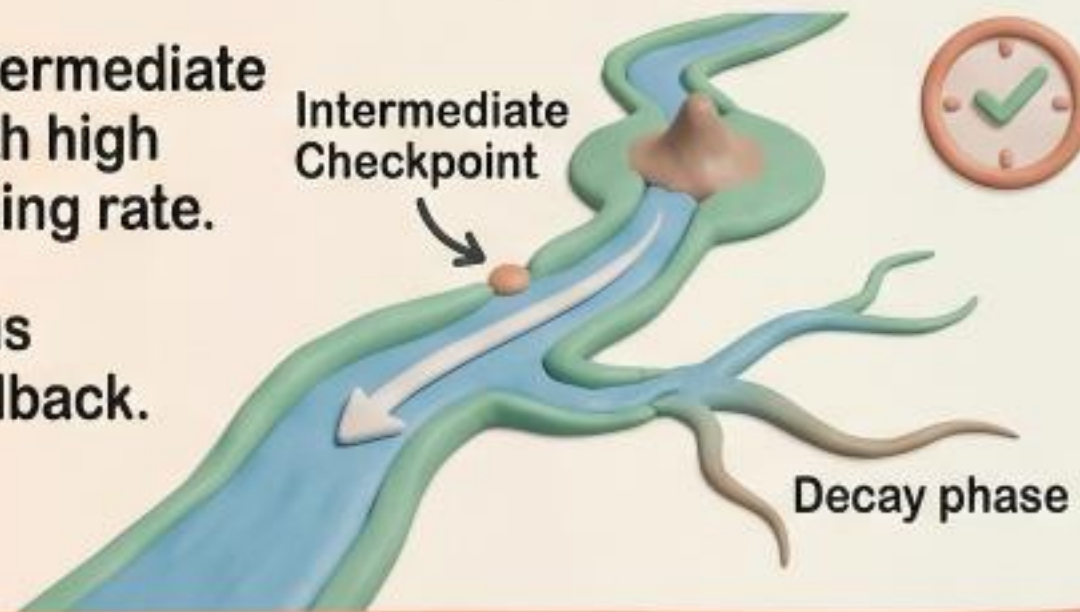
Reproduces WSD's non-traditional loss curve by aligning learning phases with the landscape's topography.

WSD-S: Simplified and Improved Version

Resumes Immediately, No Rollback

Starts from intermediate checkpoint with high constant learning rate.

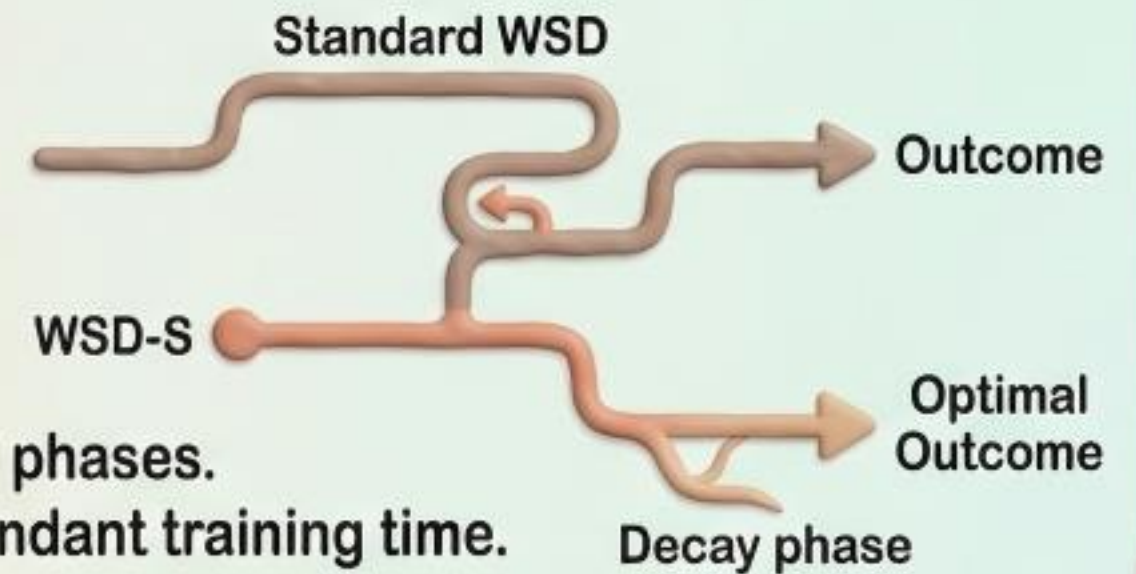
Avoids previous checkpoint rollback.



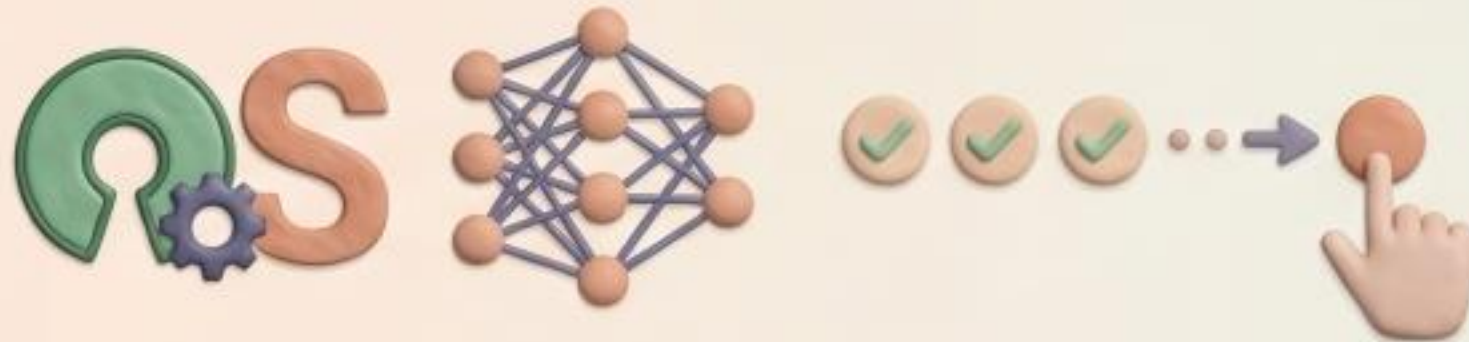
More Efficient Compute Use



Reuses decay phases.
Reduces redundant training time.



User-Friendly for Open-Source Models



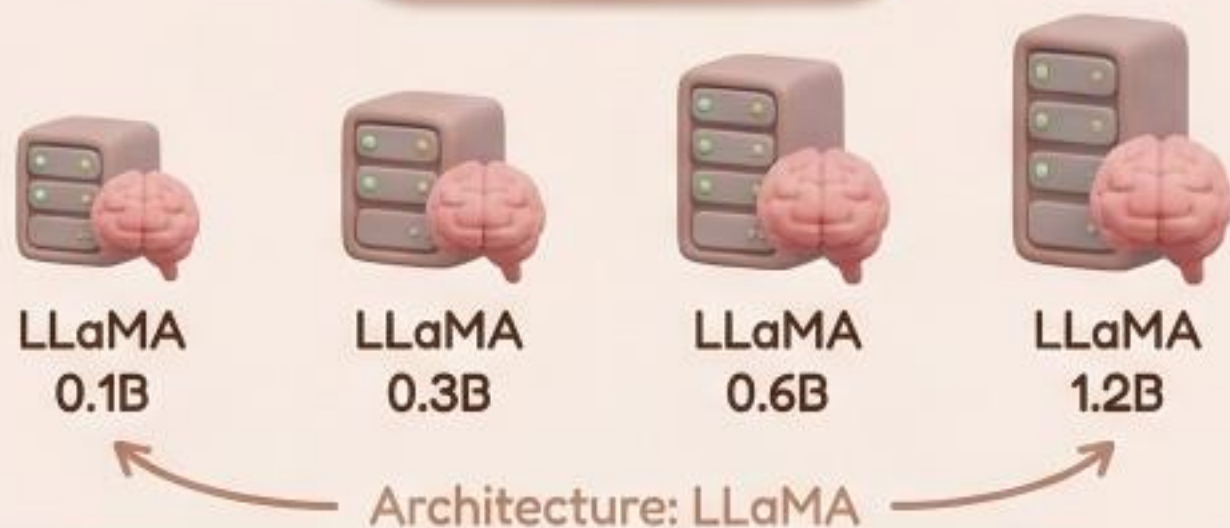
Core Benefits Overview

- Immediate Resumption: Faster training start
- Efficient Compute: Reuses existing training phases
- Open-Source Ready: Simplified model continuation



Experimental Setup

Models



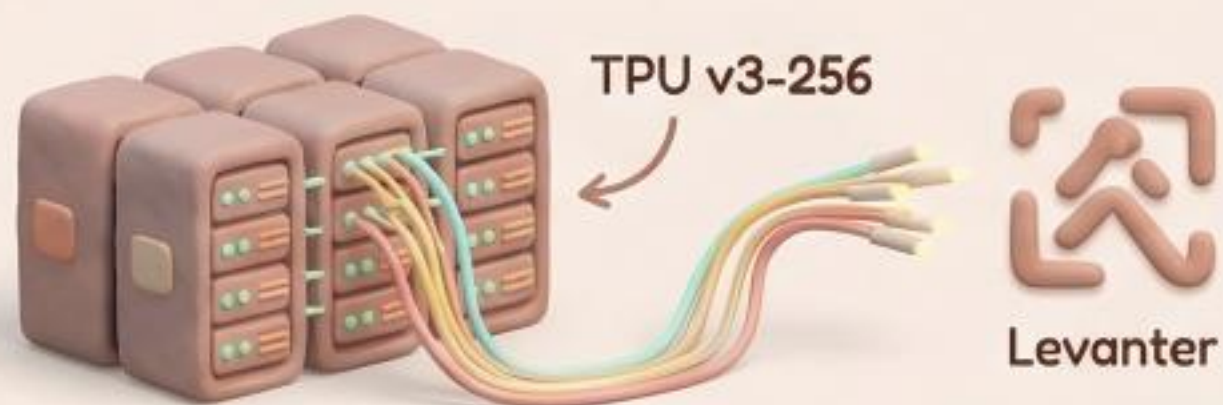
Dataset



Context Length: 4096

Batch Size: 4M Tokens

Training



Framework: Levanter

Hyperparameters

Peak LR (0.1B/0.3B): $6e-4$

Peak LR (0.6B/1.2B): $4e-4$

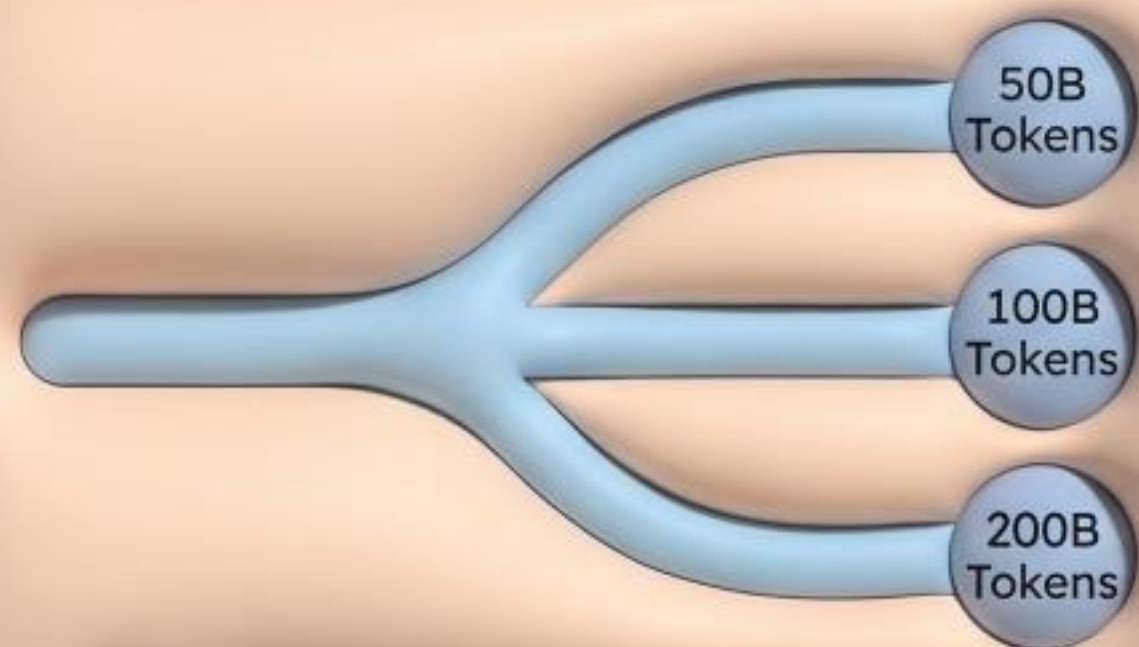
Decay Fraction **10%**

Decay: 10% of Training Steps

WSD-S vs Cosine-Oracle: Competitive Performance

WSD-S delivers competitive results compared to Cosine-Oracle in a single run, eliminating the need for separate training for each compute budget.

WSD-S: Single-Run Efficiency

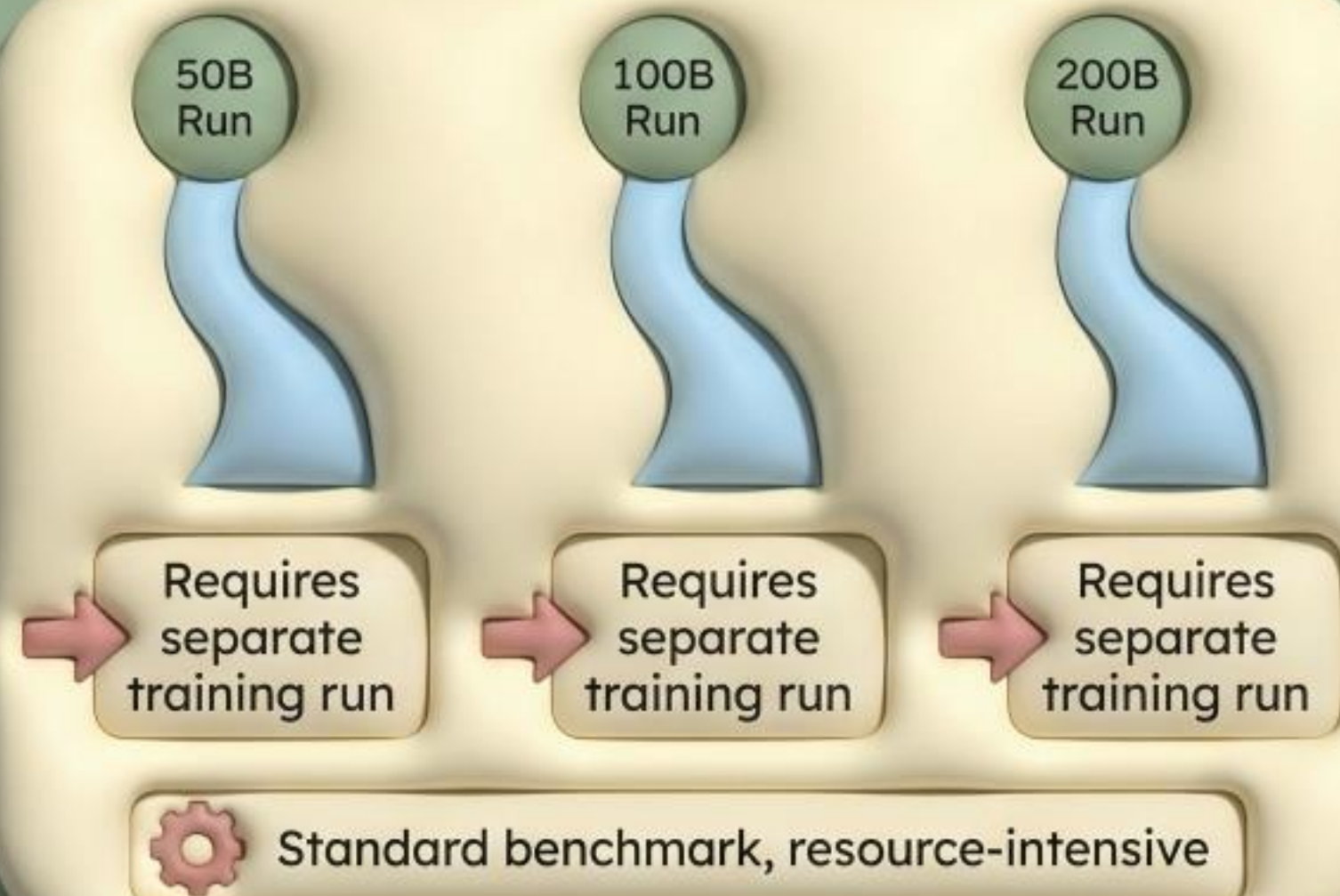


Multiple compute budgets from one run



Largest validation loss gap is only $6e-3$

Cosine-Oracle: Multi-Run Benchmark



WSD-S vs Cyclic-Cosine: Significant Outperformance

Performance Gap



Common Belief vs. Our Finding



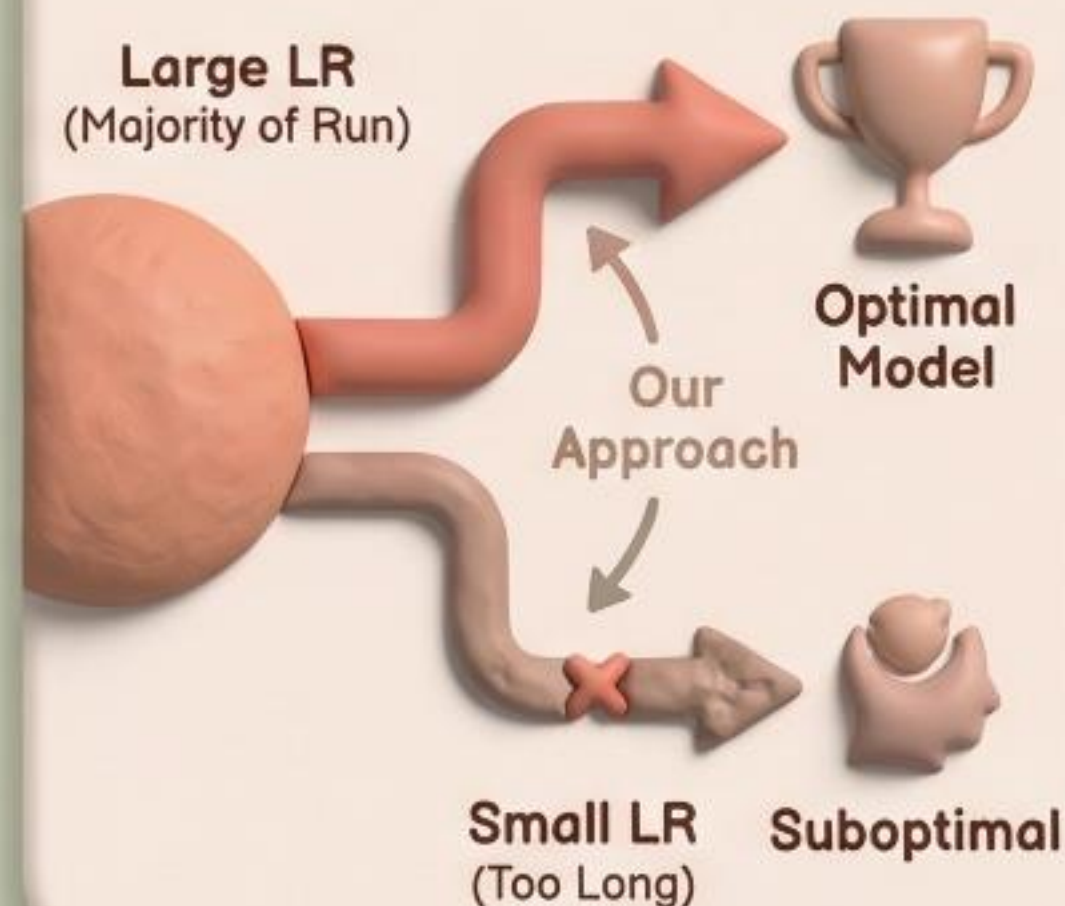
Common Belief

Loss spiking after increasing LR causes performance loss.

Our Finding

Model trained with large LR for majority of run is better. Too long on small LR implicitly hurts.

Optimal Strategy



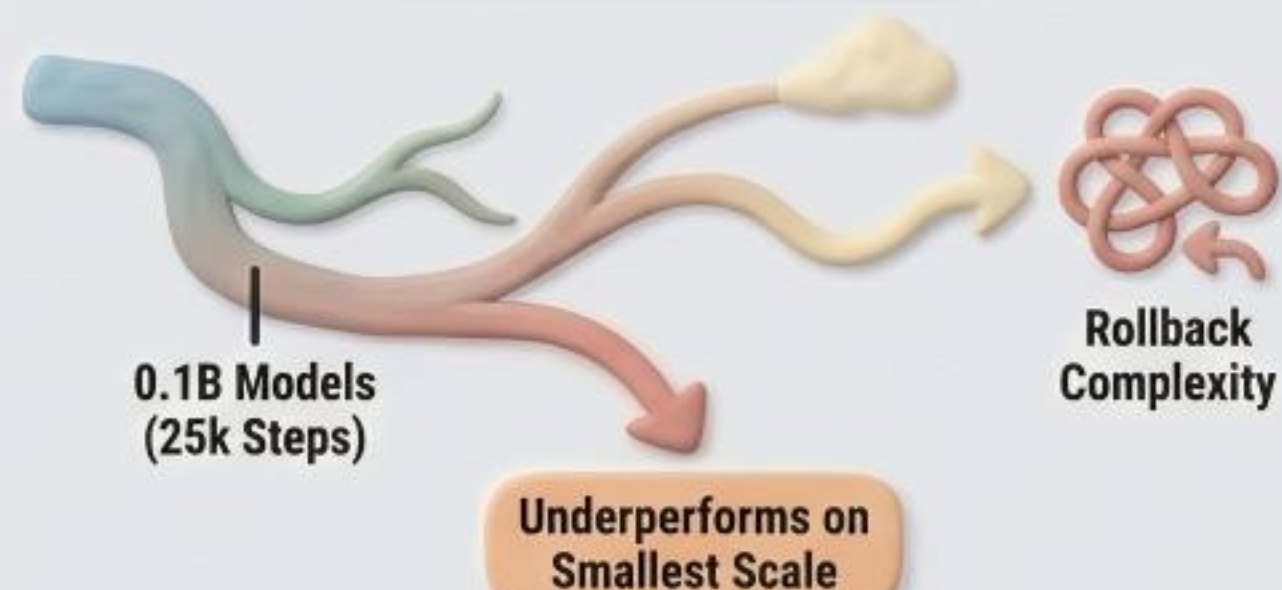
WSD-S vs WSD: Improved Efficiency

WSD-S (Improved Efficiency)



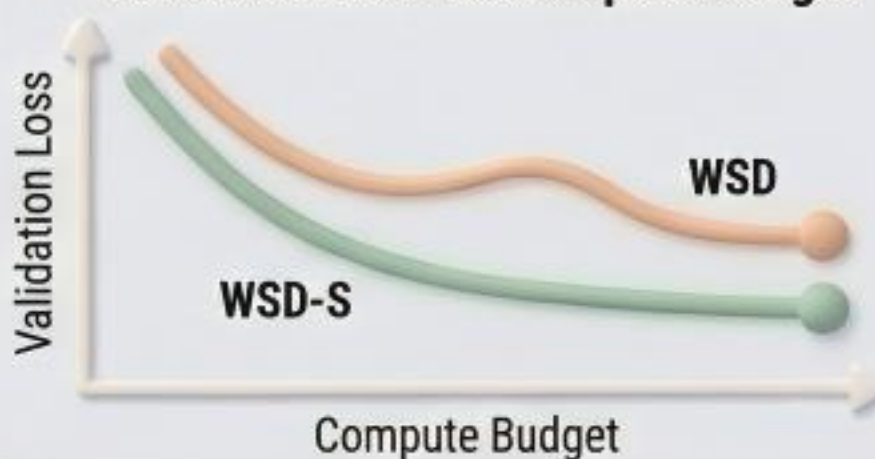
- Consistently outperforms WSD on 200B tokens across models.
- Reuses decay, achieving lower loss for same compute.
- No rollback complexity, easier for continual learning.

WSD (Standard)



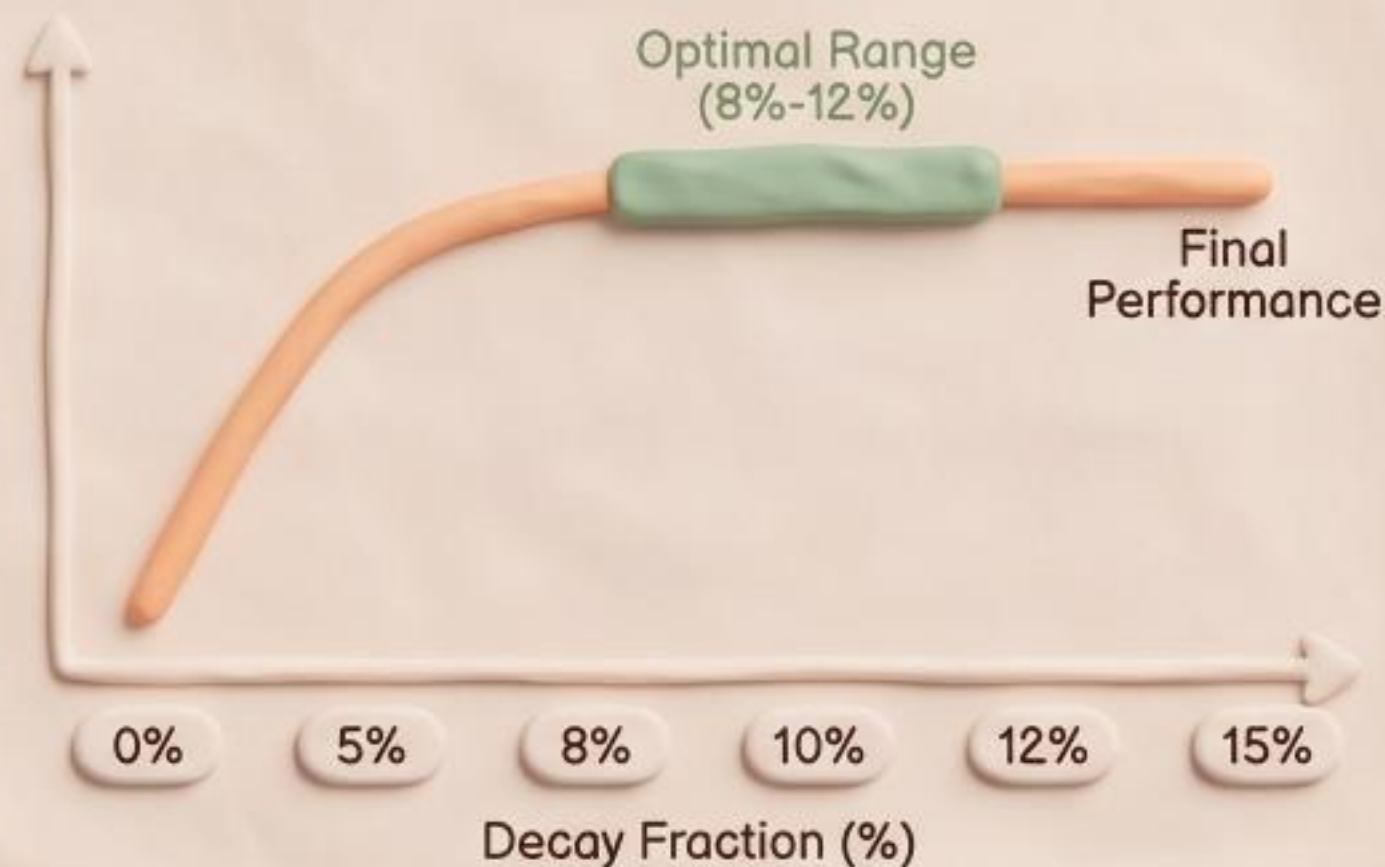
- Underperforms only on smallest scale (0.1B, 25k steps).
- Requires rollback for continual learning, adding complexity.

Validation Loss vs. Compute Budget



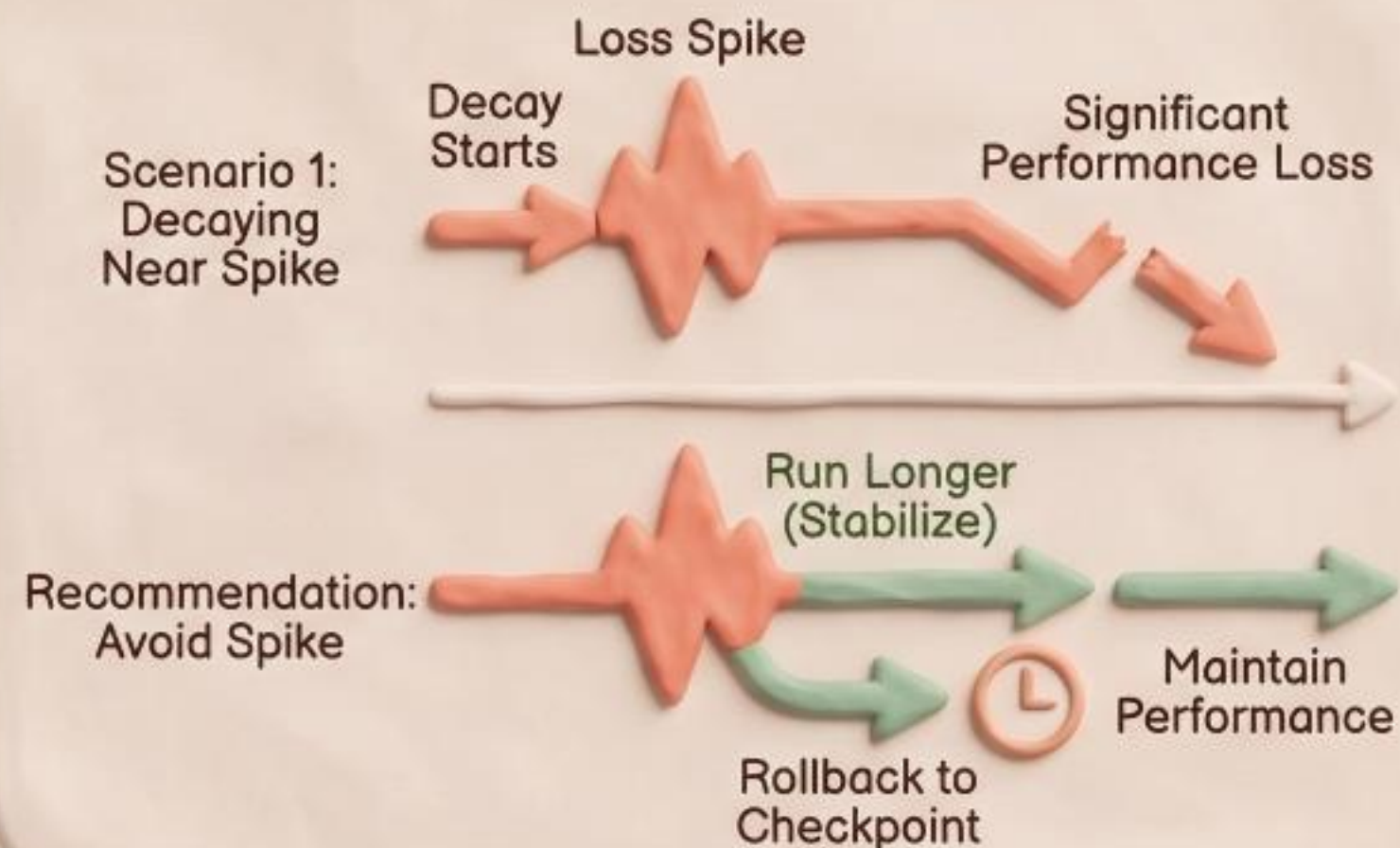
Ablation: Sensitivity to Decay Fraction

Insensitivity to Decay Fraction



Performance is relatively insensitive within the 8%-12% range, with similar final results.

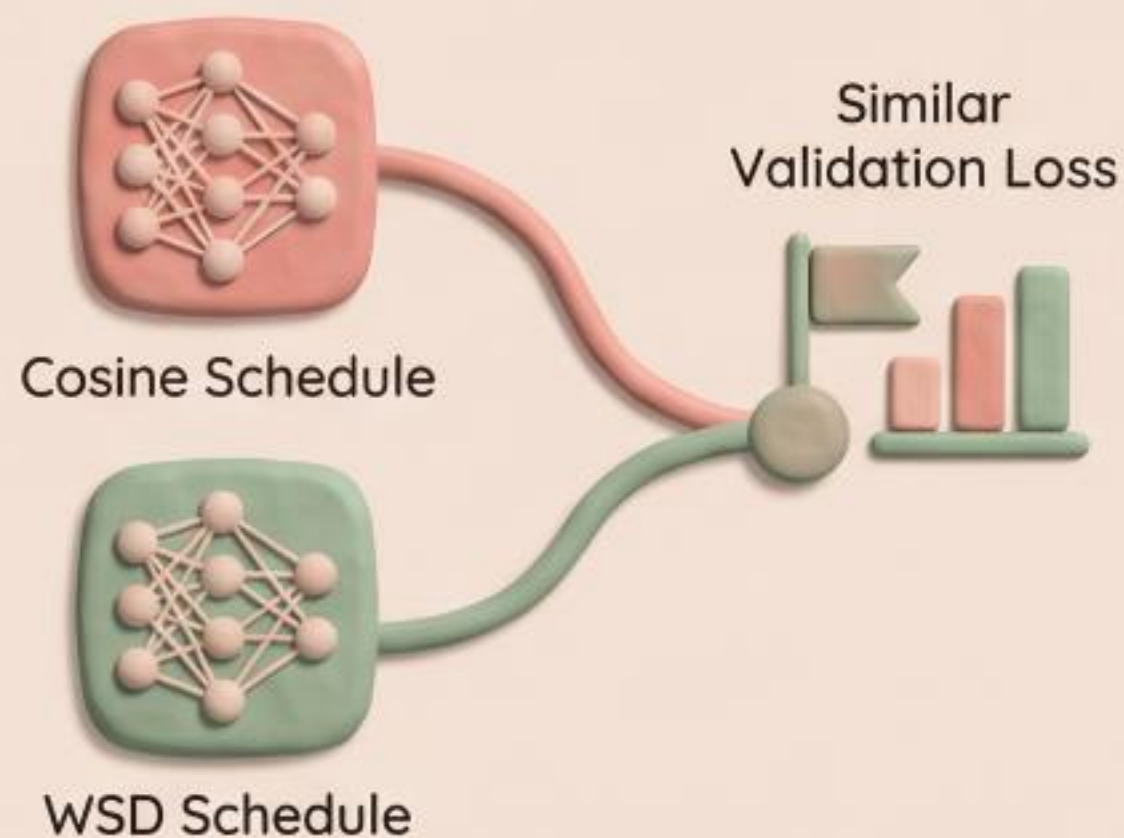
Risk: Decaying Near Loss Spike



Decaying near a loss spike can significantly degrade validation loss. Recommended: Run longer or rollback.

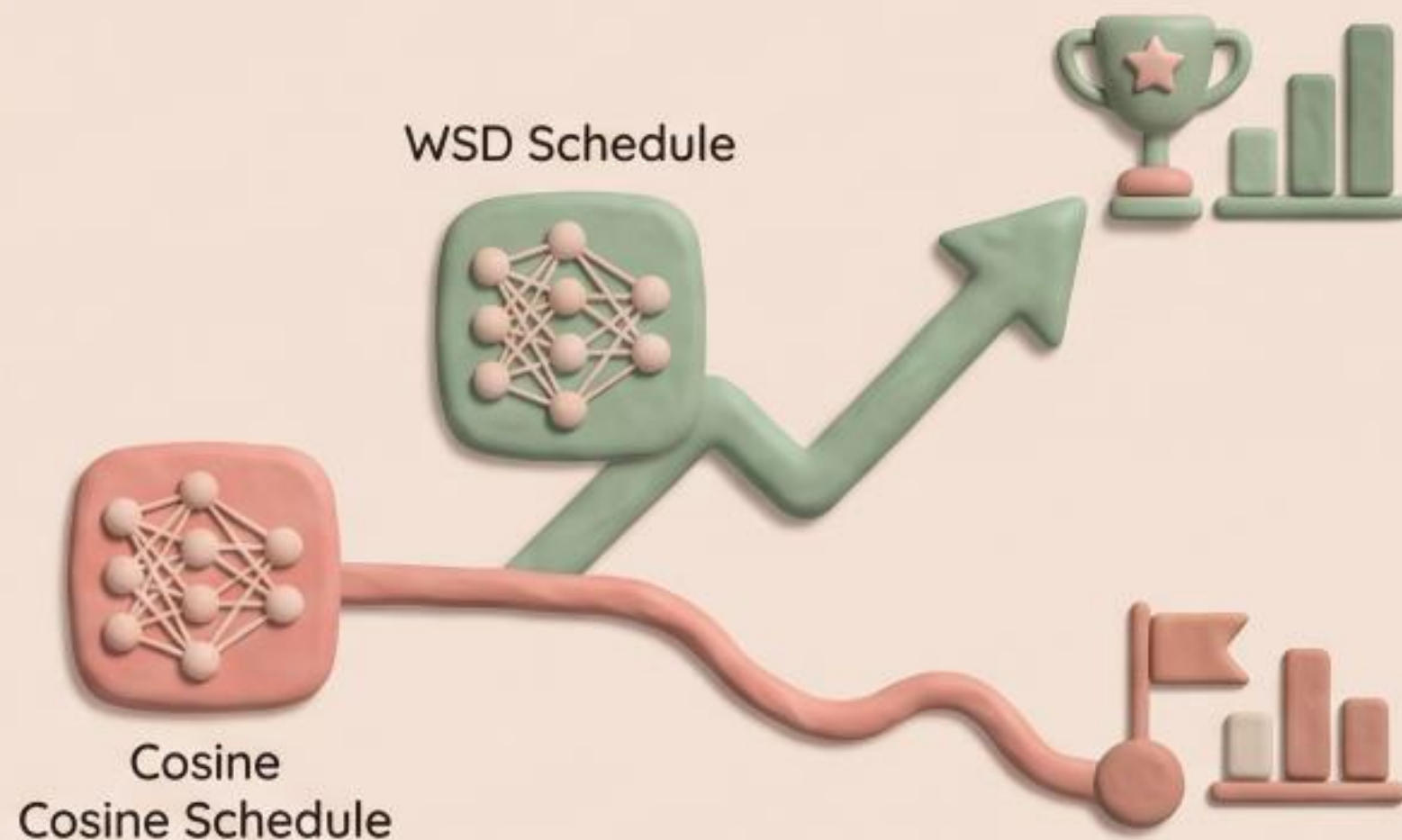
Cosine Implicitly Hurts Models for Continual Learning

Initial Training (Single Run - 50B Tokens)



Both schedules may achieve similar validation loss initially.

Continual Learning (Further 50B Tokens)



WSD-S Consistently Outperforms Cosine as a Starting Point for Further Training.

Key Insights and Contributions

1. River Valley Metaphor

- Large LR makes implicit progress, revealed only upon decay.

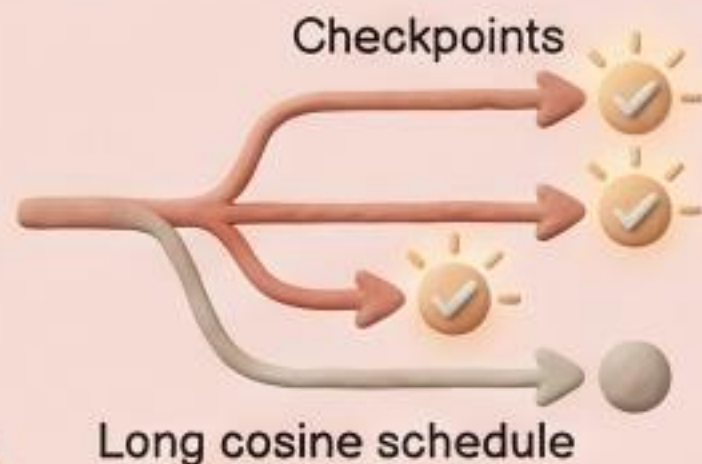


2. Simple Data Model

- Generates river valley; attributes sharpness to token uncertainty.



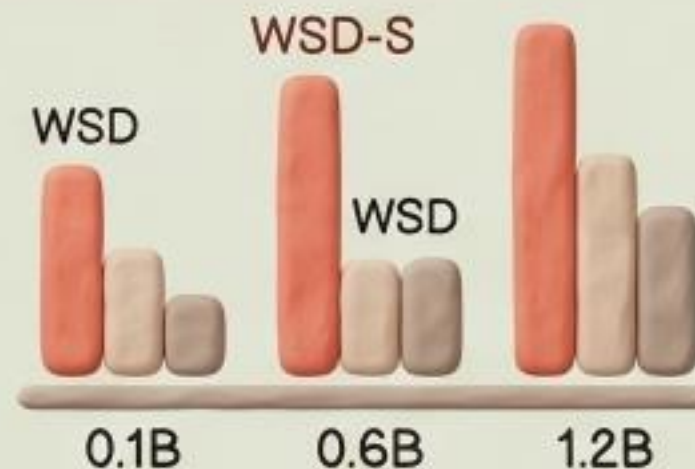
3. WSD-S Efficiency



- Single Run = Multiple Checkpoints (As good as tuned cosine)

4. WSD-S Outperformance

- Outperforms WSD & Cyclic-Cosine across 0.1B to 1.2B parameters.



Future Directions



Extend to K-dimensional

Broaden the river valley concept to multi-dimensional landscapes.



Analyze Non-straight Rivers

Investigate SGD dynamics in complex, non-linear river structures.



Improve LR Upper Bounds

Refine theoretical limits through finer analysis for better convergence.



Explore Other Algorithms

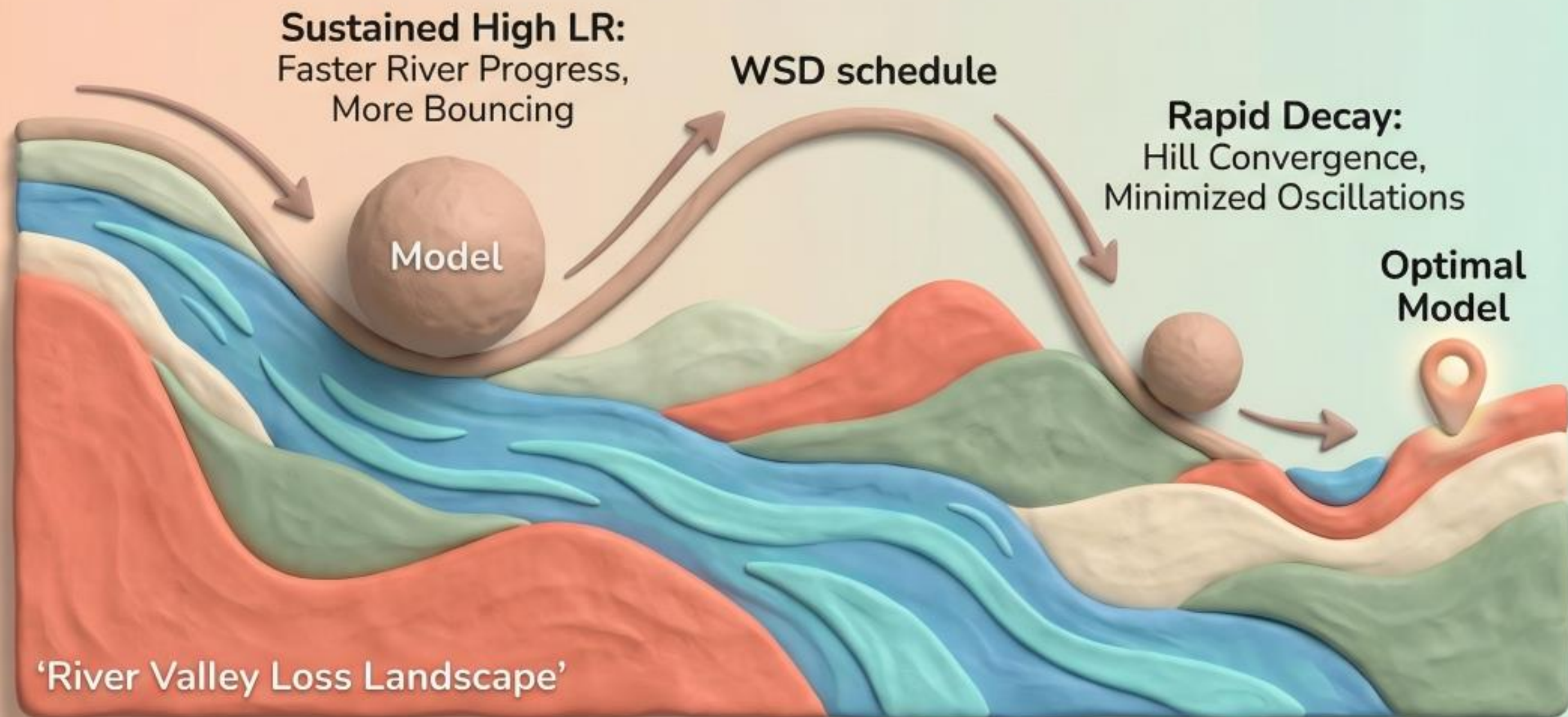
Apply river valley insights to diverse optimization techniques beyond SGD.



Investigate Other Domains

Examine river valley landscapes in fields beyond language modeling.

Conclusion



The river valley framework provides a powerful understanding of WSD's success. WSD-S offers a practical, efficient solution for continual learning in language model pretraining.